Prompt Engineering Quiz

15 Questions • 4 Topics

Topics Covered

1 **Prompt Design Principles**
Specificity, system prompts, output formatting, and core design principles.

2 **Zero-shot & Few-shot Prompting**
When and how to use zero-shot vs few-shot approaches.

3 **Chain-of-Thought Prompting**
Step-by-step reasoning, CoT triggers, and when CoT helps most.

4 **Prompt Optimization Techniques**
Iteration, role prompting, negative examples, and refinement strategies.

Prompt Design Principles

Q1. Which of the following best describes 'prompt engineering'?

A) Writing code to generate AI models
B) Crafting inputs to guide AI models toward desired outputs
C) Designing the training dataset for language models
D) Optimizing the neural network architecture

Answer: **B)** Crafting inputs to guide AI models toward desired outputs___

Q2. What does the principle of 'specificity' in prompt design mean?

A) Using technical jargon to impress the model
B) Keeping prompts as short as possible
C) Providing clear, detailed instructions that reduce ambiguity
D) Always using bullet points in your prompts

Answer: **C)** Providing clear, detailed instructions that reduce ambiguity

Q3. Which prompt component is most important for setting the model's behavior and persona?

A) The user message

B) The system prompt
C) The temperature setting
D) The stop sequence

Answer: **B)** The system prompt

Q4. Why should you specify the desired output format in your prompt?

A) It speeds up the model's inference time
B) It reduces API costs
C) It helps the model structure its response in a usable way
D) It improves the model's factual accuracy

Answer: **C)** It helps the model structure its response in a usable way

Zero-shot & Few-shot Prompting

Q5. In zero-shot prompting, how many examples are provided to the model?

A) Zero
B) One
C) Two or three
D) As many as needed

Answer: **A)** Zero

Q6. What is the primary advantage of few-shot prompting over zero-shot?

A) It always produces shorter responses
B) It demonstrates the desired pattern through examples
C) It requires no prompt engineering skills
D) It works better for math tasks only

Answer: **B)** It demonstrates the desired pattern through examples

Q7. When designing few-shot examples, which factor matters most?

A) Using as many examples as possible
B) Choosing diverse, representative, high-quality examples
C) Writing examples that are longer than the expected output
D) Copying examples from the training data

Answer: **B)** Choosing diverse, representative, high-quality examples

Q8. Which scenario is best suited for zero-shot prompting?

A) Translating text into a highly specific custom format
B) Generating a creative poem on a common topic
C) Extracting data into a precise JSON schema
D) Classifying text into domain-specific categories

Answer: **B)** Generating a creative poem on a common topic

Chain-of-Thought Prompting

Q9. What is chain-of-thought (CoT) prompting designed to improve?

A) The speed of API responses
B) The model's reasoning on complex multi-step problems
C) The length of generated outputs
D) Image generation quality

Answer: **B)** The model's reasoning on complex multi-step problems

Q10. Which phrase best activates chain-of-thought reasoning in a zero-shot CoT prompt?

A) "Give me a short answer."
B) "Answer immediately."
C) "Let's think step by step."
D) "Be concise and direct."

Answer: **C)** “Let’s think step by step.”

Q11. Few-shot chain-of-thought combines few-shot examples with CoT. What do the examples include?

A) Only the final answers
B) The question and the reasoning chain, plus the final answer
C) Just the reasoning without a final answer
D) Random unrelated examples

Answer: **B)** The question and the reasoning chain, plus the final answer

Q12. On which type of task does chain-of-thought prompting show the LEAST benefit?

A) Mathematical word problems
B) Logical deduction puzzles
C) Simple factual lookups
D) Multi-step reasoning tasks

Answer: **C)** Simple factual lookups

Prompt Optimization Techniques

Q13. What is 'prompt iteration' in the context of prompt optimization?

A) Training a new model on your prompts
B) Repeatedly refining prompts based on output quality
C) Adding more tokens to each prompt
D) Using multiple AI providers simultaneously

Answer: **B)** Repeatedly refining prompts based on output quality

Q14. Which technique involves giving the model a role or persona to improve output quality?

A) Temperature tuning

B) Role prompting
C) Stop sequencing
D) Beam search

Answer: **B)** Role prompting

Q15. What is the purpose of including negative examples (what NOT to do) in a prompt?

A) To confuse the model
B) To reduce token usage
C) To explicitly rule out unwanted patterns or behaviors
D) To improve inference speed

Answer: **C)** To explicitly rule out unwanted patterns or behaviors